

Synthetic Validation Manual

This PDF is generated from the Markdown manual for teaching rich-text rule extraction.

Rule ID	Intent	Native Pattern	Expected Evidence	Review
RESET_STABILITY	Verify reset release stability	SET RESET_N=0; WAIT 10ns; SET RESET_N=1; WAIT 20ns; CHECK READY==1	READY becomes 1	Confirm reset polarity
VOLTAGE_SWEEP	Validate voltage range	FOR VDD IN [0.9, 1.0, 1.1]: SET VDD; RUN BASIC_OP; CHECK PASS==1	PASS remains 1	Confirm voltage range
JITTER_TOLERANCE	Validate clock jitter tolerance	SET CLK_JITTER=50ps; RUN BASIC_OP; CHECK ERROR_COUNT==0	ERROR_COUNT remains zero	Confirm jitter model

Environment Adaptation Rules

- Native signal RESET_N maps to environment signal rst_n.
- Native signal READY maps to environment signal ready.
- Native signal VDD maps to environment parameter supply_vdd.
- Native signal CLK_JITTER maps to environment parameter clock_jitter_ps.
- Native macro BASIC_OP maps to environment function basic_transaction().

Human Review Gate

The reviewer must confirm polarity, legal voltage range, jitter model, and expected evidence.